

CHEN HEXIN

Beijing, P.R.China

<https://github.com/JJsoga> | hx-chen24@mails.tsinghua.edu.cn

EDUCATION

Tsinghua University, China

09/2024–06/2028

B.S. in Mathematics and Physics; B.E. / B.S. in Software Engineering (Dual Degree)

GPA: 3.8/4.0 (Ranking Top 15%)

Coursework: Calculus, Linear Algebra, Probability and Stochastic Progress, Discrete Mathematics I-II, Data Structures, Fundamental Programming, Principles of Database Systems, Computer Organization Principle

RESEARCH EXPERIENCE

Cognitive-Breath: Cognitive Load Sensing through Eyeglass Nose Pads

Research Assistant; University of Chicago

06/2026–09/2026

- Developed an end-to-end pipeline for cognitive-load sensing from breathing signals captured by contact microphones embedded beneath eyeglass nose pads, spanning signal preprocessing, feature engineering, model development, cross-user evaluation, and real-time inference.
- Processed acoustic signals using 150–1000 Hz Butterworth band-pass filtering, 64 ms framing, and log root-mean-square (RMS) envelope extraction; engineered temporal-spectral features capturing breathing rhythm, signal intensity, cycle morphology, and bilateral-channel synchrony.
- Built lightweight classifiers using regularized logistic regression and gradient boosting, and introduced causal online mean/variance normalization with participant-specific calibration to mitigate cross-user distribution shifts, improving best block-level classification accuracy by approximately 17% relative to the baseline.
- Conducted leave-one-participant-out cross-user evaluation, feature ablation, bilateral-channel fusion, and robustness analyses under acoustic and motion interference; also identified sampling-rate drift in synchronized acoustic and nasal-pressure recordings to improve temporal alignment and validation.
- Implemented a streaming inference and visualization prototype that continuously processed live sensor data and updated cognitive-load estimates every 5 seconds while preserving causal, calibration-free inference for unseen users.

Publication: Co-first-author manuscript currently under review.

Audio Signal Representation Project

Research Assistant; Advisor: Dr. Jiangyan Yi, Tsinghua University

06/2025 - 03/2026

- Worked on a neural audio representation pipeline for academic research and prototype-level experiments.
- Refactored the quantization module to improve modularity, reconstruction stability, and downstream usability.
- Implemented and evaluated alternative vector-quantization designs for separating global audio patterns from local acoustic details.
- Ran controlled experiments and ablation studies to compare system components.

Motion-Robust IMU Signal Processing for Wearable Interaction

Research Assistant; Tsinghua University

09/2024–07/2025

- Analyzed multi-axis accelerometer signals in both time and frequency domains, identifying noise patterns and non-stationary motion artifacts in fine-grained wearable sensing data.
- Developed an adaptive signal filtering pipeline to reduce motion-induced jitter while preserving subtle intentional movement cues under low-latency constraints.
- Improved trajectory stability and repeatability in prototype experiments with minimal impact on system responsiveness.

LEADERSHIP & ACTIVITIES

Team Captain, “Qinglai Zhixing” Social Practice Team, Tsinghua University	07/2025–08/2025
Team Lead, Tsinghua–CUC Game Jam	12/2024
Group Lead, Student Practice Committee, Wuqiong College, Tsinghua University	08/2025–Present
Contestant, MCM/ICM Mathematical Modeling Contest	02/2025

HONORS & AWARDS

Academic Excellence Scholarship	10/2025
Merit Scholarship	10/2025
Silver Award, Summer Social Practice Team, Tsinghua University	12/2025
MCM/ICM Mathematical Modeling Contest, Successful Participant Award	01/2025

CORE SKILLS

Languages: Mandarin Chinese (Native); English (TOEFL 100)

Mathematics: Calculus, Linear Algebra, Probability & Stochastic Processes, Discrete Mathematics, Mathematical Modeling, Statistical Analysis

Programming & Engineering: Python, C++, C, Java, Arduino; Signal Processing, Databases, Computer Systems, Software Engineering

Machine Learning & Research: Machine Learning, Time-Series Analysis, Feature Engineering, Model Training & Evaluation, Controlled Experiments, Statistical Analysis

Leadership & Collaboration: Team Leadership, Cross-functional Collaboration, Project Coordination, External Outreach